Junghun Hwang (황정헌)

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Education

Konkuk University

Mar 2022 - Expected Feb 2028
B.S. in Smart Vehicle Engineering | Seoul, Korea
In progress: AI Mobility Systems

Research Interests

I am interested in perception and decision-making for autonomous driving, especially learning-based and end-to-end driving.

Research & Project Experience

Kookmin Autonomous Driving Competition - AI-Edge Challenge

Apr 2026 - Aug 2026
Team Lead | Team of 5

- Built a ROS 2 imitation-learning pipeline with ResNet18 and a LiDAR 1D CNN. Collected driving data, trained models, deployed TorchScript inference, and enabled switching between learned and rule-based control.
- Prepared and reviewed datasets in Roboflow, trained YOLOv8n and YOLO11n object detectors, and selected a model based on validation mAP50 and mAP50–95.
- Integrated ROS 2 mission-management and driving modules, adjusted lane-following parameters, and configured mission transitions on the competition vehicle. The team used rule-based control in the final competition.
- Defined parking waypoints on a SLAM map and implemented reverse recovery logic for the parking event.
- Examined learned steering in simulation and vehicle trials, identifying training-data coverage, data quality, and perception–decision–control integration as priorities for vehicle operation.

Team results: **Preliminary 2nd Place** | Finals 13th Place (course completed) | **Parking Special Award**

2026 Student AI · SW Mobility Competition

Mar 2026 - Oct 31, 2026 (scheduled end)
Team of 27 | Decision-Making Subteam (5)

- Designed FSM states and transition conditions with teammates for ROS 2 behavior planning, covering traffic-light handling, timed stops, obstacle avoidance, and mission completion.
- Recorded GNSS coordinates while driving to establish route waypoints and conducted vehicle trials using GNSS/IMU-based localization. Completed waypoint-following runs on both preliminary and final courses at speeds up to **11 m/s (39.6 km/h)** during venue practice.
- Traced perception outputs, FSM transitions, planned paths, and vehicle state in Foxglove, checking timestamps and coordinate frames. Reviewed recordings with teammates to identify issues and decide corrections; used driving data to make debugging more systematic.
- Operated the vehicle during obstacle-avoidance field tests and confirmed deceleration before avoidance, collision-free passage, return to the reference route, and subsequent acceleration.
- Presented a Pure Pursuit–Stanley hybrid strategy at a symposium sponsored by Hyundai Mobis, then implemented steering blending by driving phase in ROS 2.
- Compared hybrid tracking with Stanley alone on matched Gazebo course segments at 6 m/s; the initial comparison showed approximately **14%** lower 95th-percentile heading error and **10%** lower total variation in steering commands.

Camera-Based V2I Cooperative End-to-End Autonomous Driving

2026 - Ongoing
Capstone Design Project | Team of 5 | Research & Design Phase

- Reviewed UniV2X's vehicle–infrastructure fusion framework and the scope of its open-loop evaluation. Wrote a literature summary to guide project scoping and a planned small-scale closed-loop comparison.
- Prepared the midterm slides after team design discussions. Described the planned E2E comparison, pedestrian-risk assessment, and target-speed labels based on braking margins.
- Investigating V2I communication options, including ROS 2/DDS over Wi-Fi. The review covers message freshness, latency, and reception continuity.

UAV Waypoint Mission - ROS 2 / PX4 SITL

May 2026 - Jun 2026
Autonomous System Platform Course | UAV Waypoint Development | Team of 4

- Developed a ROS 2 mission package with CSV waypoints, TF-based arrival checks, and coordinate-mode handling; integrated it with an existing PX4 offboard controller via PoseStamped commands.
- Demonstrated sequential waypoint tracking and final autonomous landing in the PX4 SITL/Gazebo course simulation.

ROS 2 / CARLA Decision and Control

Mar 2026 - Jun 2026
Dream Semester Project | Decision-Making | Team of 3

- Implemented ROS 2 decision-making, Pure Pursuit waypoint tracking, and longitudinal control with CARLA integration and later vehicle-interface adaptation.
- Investigated steering behavior in CARLA by adjusting steering direction, lookahead, smoothing, and command limits.

Awards & Competitions

2026 Kookmin AI-Edge Challenge
Parking Special Award (team)

2026 Hyundai Mobis Technical Special Award
Autonomous Driving Division | Team K.A.I. (team award)
Feb 10, 2026 | Oral presentation, jointly credited to Woojin Hwang and Junghun Hwang: *Decision Logic: Combining Pure Pursuit and Stanley for Scenario-Adaptive Control*.

2022 Drone Clash, General Division
3rd Place

Technical Skills

Programming: Python, C++
Autonomy & Control: ROS 2, FSM-based behavior planning, Pure Pursuit–Stanley tracking
Learning & Perception: PyTorch, imitation learning, YOLO, TorchScript, OpenCV, Roboflow
Simulation & UAV: CARLA, Gazebo, PX4 SITL, QGroundControl
Data & Development: ROS 2 log analysis, Foxglove, rosbag2 / MCAP, RViz2, Ubuntu, Git/GitHub